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Chennai-602105

**PCA-Based District-Wise Drinking Water Quality Risk Index and Disease
Outbreak Correlation System**

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfilment for the Course of

CSA0904 - PROGRAMMING IN JAVA

to the award of the degree of

BACHELOR OF ENGINEERING

IN

INFORMATION TECHNOLOGY

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September 2026



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DECLARATION

We, **K. Anjali Giri (192421037), M. Moshika (192424064), Rose Mystica. J (192421410)** of the **[Information Technology]**, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled “**PCA-Based District-Wise Drinking Water Quality Risk Index and Disease Outbreak Correlation System**” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled **“PCA-Based District-Wise Drinking Water Quality Risk Index and Disease Outbreak Correlation System”** has been carried out by **M. Moshika (192424064), Anjali Giri .K (192421037), Rose Mystica .J (192421410)** under the supervision of **Dr. R. Thalpathi Rajasekaran** and is submitted in partial fulfilment of the requirements for the current semester of the **B.E computer science engineering** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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ACKNOWLEDGEMENT

We would like to express our heartfelt gratitude to all those who supported and guided us throughout the successful completion of our Capstone Project. We are deeply thankful to our respected Founder and Chancellor, Dr. N.M. Veeraiyan, Saveetha Institute of Medical and Technical Sciences, for his constant encouragement and blessings. We also express our sincere thanks to our Pro-Chancellor, Dr. Deepak Nallaswamy Veeraiyan, and our Vice-Chancellor, Dr. S. Suresh Kumar, for their visionary leadership and moral support during the course of this project.

We are truly grateful to our Director, Dr. Ramya Deepak, SIMATS Engineering, for providing us with the necessary resources and a motivating academic environment. Our special thanks to our Principal, Dr. B. Ramesh for granting us access to the institute's facilities and encouraging us throughout the process. We sincerely thank our Head of the Department, Dr Sasi Rekha for his continuous support, valuable guidance, and constant motivation.

We are especially indebted to our guides, Dr. R. Thalpathi Rajasekaran for their creative suggestions, consistent feedback, and unwavering support during each stage of the project. We also express our gratitude to the Project Coordinators, Review Panel Members (Internal and External), and the entire faculty team for their constructive feedback and valuable inputs that helped improve the quality of our work. Finally, we thank all faculty members, lab technicians, our parents, and friends for their continuous encouragement and support.

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ABSTRACT

Drinking water quality is an important public health concern because contaminated water can contain physical, chemical, and biological pollutants that may be associated with water-related diseases. Water quality can vary across districts due to differences in water sources, pollution, sanitation, agricultural activities, industrial discharge, and environmental conditions. Analyzing multiple water-quality parameters separately can make it difficult to identify the overall risk level of each district.

The proposed project, **“PCA-Based District-Wise Drinking Water Quality Risk Index and Disease Outbreak Correlation System,”** aims to develop a data-driven approach for assessing drinking water quality and examining its association with reported disease cases. The system uses **Principal Component Analysis (PCA)** to analyze multiple water-quality parameters and identify the major components contributing to water-quality variation. These components are used to generate a **Drinking Water Quality Risk Index (QRI)** for each district. Based on the calculated index, districts are classified into different risk categories such as Low, Medium, and High Risk.

The system further integrates disease-related data and performs **correlation analysis** between the QRI and reported disease cases. This helps identify district-wise patterns and possible associations between water-quality risk and disease occurrence. A decision-support module presents the results using risk classifications, district rankings, graphs, and visualizations.

The expected outcome is a structured system that simplifies complex water-quality data, provides district-wise risk assessment, and supports environmental and public-health monitoring. The correlation results are used to identify associations and do not establish direct causation.

TABLE OF CONTENTS

CHAPTER	CONTENT	PAGE NO
	ABSTRACT	5
1	Introduction	8-10
	1.1 Background Information	
	1.2 Project Objectives	
	1.3 Significance	
	1.4 Scope	
	1.5 Methodology Overview	
2	Problem Identification and Analysis	11-12
	2.1 Description of the problem	
	2.2 Evidence of the problem	
	2.3 Working Principles	
	2.4 Supporting Data/Research	
3	Solution Design And Implementation	13-15
	3.1 Development and Design Process	
	3.2 Tools And Technologies Used	
	3.3 Solution Overview	
	3.4 Engineering Standards Applied	
	3.5 Solution Justification	
4	Result and Recommendations	16-17
	4.1 Evaluation of Results	
	4.2 Challenges Encountered	
	4.3 Possible Improvements	
	4.4 Recommendations	
5	Reflecting Learning and Personal Development	18-21
	5.1 Key Learning Outcomes	
	5.2 Academic Knowledge	

	5.3 Technical Skills	
	5.4 Problem-Solving and Critical Thinking	
	5.5 Conclusion of Personal Development	
6	Conclusions	22-23
7	References	24
8	Appendices	

LIST OF THE TABLES

Table No.	Title	Page No
1.1	Important Drinking Water Quality Parameters	8
1.3	Major Contribution and its Purpose	13
3.1	Tools and Technologies	15
4.1	Evaluation Parameters and Their Purpose	23

LIST OF FIGURES

Figure No.	Title	Page No
2.2	Drinking Water Quality Testing and Analysis	8
3.3	System Architecture Diagram	23

CHAPTER 1

INTRODUCTION

1.1 Background Information

Drinking water quality is an important factor in maintaining public health. Water contains different physical and chemical parameters such as pH, turbidity, total dissolved solids, hardness, fluoride, nitrate, chloride, and iron. When these parameters exceed acceptable levels, the water may become unsuitable for consumption and may be associated with health risks.

The proposed system uses **Principal Component Analysis (PCA)** to analyze multiple water-quality parameters and identify the major factors contributing to variations in water quality. Based on these components, a **District-Wise Drinking Water Quality Risk Index** is developed. The calculated risk levels are then compared with available disease outbreak data using correlation analysis.

Category	Parameters
Physical	pH, Turbidity, TDS
Chemical	Hardness, Fluoride, Nitrate, Chloride, Iron
Biological	Total Coliform

Table 1.1: Important Drinking Water Quality Parameters

1.2 Project Objectives

The main objective of the project is to develop a district-wise system for assessing drinking water quality risk using PCA and analyzing its relationship with reported disease cases.

Key Objectives

1. To collect and preprocess district-wise drinking water quality data.
2. To analyze important water-quality parameters using PCA.
3. To identify the major components influencing water-quality variation.
4. To develop a **District-Wise Water Quality Risk Index**.
5. To compare water-quality risk with reported disease cases.
6. To perform correlation analysis between water-quality risk and disease occurrence.

7. To present the results using tables, graphs, and visualizations.

1.3 Significance

The project is significant because it provides a systematic approach for analyzing large amounts of drinking water quality data. Multiple water-quality parameters can make direct analysis difficult, so PCA helps identify the most influential components and simplify the dataset.

The developed risk index provides a clear way to compare water-quality conditions between districts. The correlation analysis adds another level of analysis by examining whether districts with higher water-quality risk also show higher reported disease cases.

The major contributions of the project are:

Contribution	Purpose
PCA Analysis	Identifies important water-quality components
Risk Index	Provides a district-wise water-quality risk measure
Risk Classification	Groups districts into different risk levels
Correlation Analysis	Examines the association between water risk and disease cases
Visualization	Makes district-wise results easier to understand

1.3: Major Contribution and its Purpose

1.4 Scope

The project focuses on the analysis of district-wise drinking water quality and its association with reported disease data. It includes data preprocessing, PCA-based analysis, risk-index development, risk classification, and correlation analysis.

Included in the Scope

- District-wise drinking water quality data.
- Preprocessing and cleaning of datasets.
- Analysis of physical, chemical, and biological parameters.
- Development of a water-quality risk index.
- District-wise risk classification.

- Graphical and tabular presentation of results.

1.5 Methodology Overview

The proposed system follows a series of steps to analyze drinking water quality and its relationship with disease occurrence.

Step 1 – Data Collection:

District-wise water-quality data and corresponding disease outbreak or disease case data are collected from suitable datasets.

Step 2 – Data Preprocessing:

The collected data is cleaned by handling missing values, removing duplicate records, correcting inconsistent values, and preparing the data for analysis.

Step 3 – PCA Analysis:

PCA is applied to the selected water-quality parameters. The analysis identifies the major components that explain the variation in the water-quality dataset.

Step 4 – Risk Index Development:

The important PCA components are used to calculate a **Water Quality Risk Index** for each district.

Step 5 – Risk Classification:

Districts are grouped into categories such as **Low Risk, Medium Risk, and High Risk** based on their calculated risk scores.

Step 6 – Disease Correlation Analysis:

The district-wise risk index is compared with reported disease cases. Correlation analysis is performed to determine the strength and direction of the association.

Step 7 – Result Visualization:

The final results are presented through graphs, tables, and district-wise comparisons to make the findings easier to interpret.

CHAPTER 2

PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the Problem

Safe drinking water is essential for public health, but its quality can vary between districts due to pollution, industrial activities, agricultural runoff, poor sanitation, wastewater, and natural environmental conditions. Water can contain physical, chemical, and biological contaminants that may affect its safety. Water quality is commonly assessed using parameters such as **pH**, **turbidity**, **Total Dissolved Solids (TDS)**, **hardness**, **fluoride**, **nitrate**, **chloride**, **iron**, and **coliform** levels.

Analyzing many parameters separately can make it difficult to understand the overall condition of water. The proposed system uses **Principal Component Analysis (PCA)** to reduce the complexity of water-quality data and identify major components. These components are used to develop a **District-Wise Drinking Water Quality Risk Index**.

2.2 Evidence of the Problem

Drinking-water contamination is an important public-health concern. The World Health Organization (WHO) reports that contaminated drinking water can transmit diseases such as diarrhoea, cholera, dysentery, typhoid, hepatitis A, and polio.

Water contamination can result from untreated wastewater, agricultural activities, industrial discharge, and poor sanitation. Different districts may therefore experience different water-quality conditions.



Figure 2.2: Drinking Water Quality Testing and Analysis

2.3 Stakeholders

Stakeholder	Role
Public Health Departments	Monitor disease patterns
Water Supply Authorities	Monitor water quality
Local Government	Support water and sanitation planning
Environmental Agencies	Monitor pollution
Healthcare Organizations	Record disease cases
Researchers	Analyze environmental and health data
General Public	Depends on safe drinking water

Table 2.1: Stakeholders and Their Roles

2.4 Supporting Data/Research

WHO provides guidelines for drinking-water quality and emphasizes risk-based monitoring to protect public health. PCA is useful for analyzing datasets containing multiple related water-quality parameters because it reduces complex variables into important components.

The proposed system combines PCA-based water-quality analysis with disease data correlation to identify district-wise risk patterns. **Correlation indicates an association and does not prove that poor water quality directly caused a disease outbreak.** The results can support further water-quality monitoring, public-health analysis, and research.

CHAPTER 3

SOLUTION DESIGN AND IMPLEMENTATION

3.1 Development and Design Process

The system follows a structured data-analysis workflow. First, water-quality and disease records are collected and matched using district and, where available, time information. Next, the water-quality dataset is cleaned and standardized. PCA is then applied to transform the original correlated indicators into a smaller set of components. Component scores are combined into a risk index. Finally, the index is compared with disease-outbreak measures using a correlation statistic and visualized for interpretation.

3.2 Tools and Technologies

Tool / Technology	Purpose
Python	Main programming language
Pandas, NumPy	Data processing and calculations
Scikit-learn	PCA and preprocessing
SciPy	Statistical and correlation analysis
Matplotlib, Seaborn	Graphs and visualization
Jupyter / Google Colab	Development and testing
Streamlit	Decision-support dashboard

Table 3.2 – Tools and Technologies

3.3 Solution Overview

Module 1: Drinking Water QRI Generation analyzes parameters such as pH, turbidity, TDS, hardness, fluoride, nitrate, chloride, iron, and total coliform. PCA identifies the major components, which are used to generate a district-wise QRI and classify districts into Low, Medium, and High Risk.

Module 2: Disease Outbreak Correlation and Risk Prediction combines the QRI with reported disease data. Correlation analysis is performed to identify associations between

water-quality risk and disease occurrence. The result provides a risk indication for further monitoring and does not represent medical diagnosis or direct causation.

Module 3: Decision Support and Risk Assessment presents QRI scores, risk categories, disease patterns, district rankings, and graphs through a simple dashboard. This helps users identify districts that may require greater attention.

System Architecture

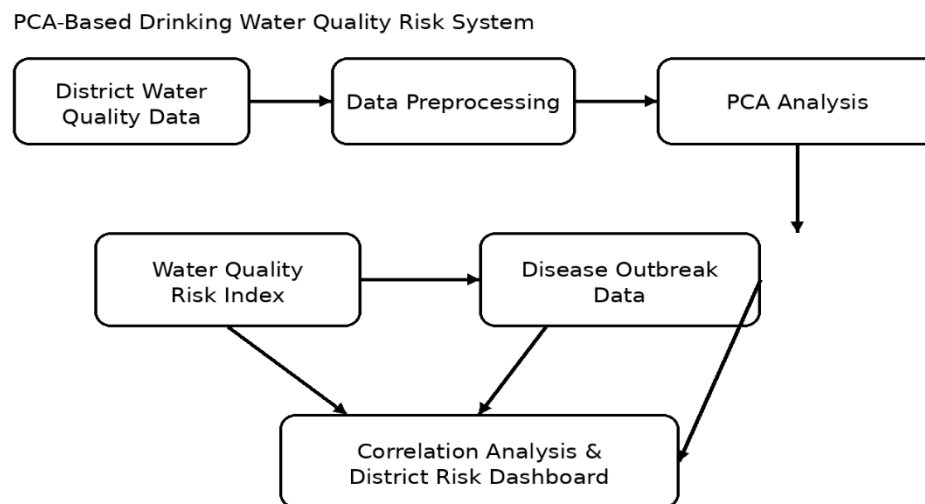


Figure 3.3 – System Architecture Diagram

3.4 Engineering Standards Applied

The project follows relevant standards to improve the reliability of both the water-quality analysis and the software system.

IS 10500:2012 – Drinking Water Specification is relevant to the project because it specifies requirements and methods of sampling and testing for drinking water in India. The standard was reviewed by BIS in 2023. The project can use the relevant water-quality requirements as a reference when interpreting water-quality parameters.

ISO/IEC 25010:2023 provides a software product quality model that can be used to evaluate characteristics of software quality. In this project, it can guide attention to aspects such as reliability, usability, performance, and maintainability.

ISO/IEC 27001:2022 provides requirements for information-security management. Where the system stores disease or other sensitive datasets, its principles can guide access control, data protection, and secure handling of information.

3.5 Solution Justification

The proposed solution is suitable because it combines **statistical analysis, water-quality assessment, disease correlation, and decision support** in a single system. PCA reduces the complexity of multiple water-quality parameters and helps identify the major patterns in the dataset. The QRI converts these results into an easier district-wise risk measure.

Using the relevant drinking-water standard provides a reference for interpreting water-quality conditions, while software-quality principles help ensure that the developed system is reliable and usable. Security principles are useful when handling datasets containing sensitive information.

Overall, the proposed design provides a structured approach from **raw data collection to final risk assessment**, making the results easier to analyze, visualize, and use for further environmental and public-health monitoring.

CHAPTER 4

RESULTS AND RECOMMENDATIONS

4.1 Evaluation of Results

The proposed system provides a systematic approach for analyzing district-wise drinking water quality and its association with reported disease cases. In **Module 1**, PCA reduces the complexity of multiple water-quality parameters and identifies the major components contributing to water-quality variation. The resulting **Drinking Water Quality Risk Index (QRI)** provides a score for each district and classifies districts into Low, Medium, and High Risk categories.

In **Module 2**, the QRI is compared with reported disease cases using correlation analysis. This helps identify the strength and direction of the association between water-quality risk and disease occurrence. **Module 3** presents the results through graphs, district rankings, and risk categories, making the information easier to understand and use for monitoring.

Expected Output Parameters

Output Parameter	Purpose
QRI Score	Measures district-wise water-quality risk
Risk Level	Classifies districts as Low, Medium, or High Risk
PCA Components	Identifies major water-quality factors
Correlation Value	Shows association between QRI and disease cases
District Ranking	Identifies districts requiring greater attention

Table 4.1: Evaluation Parameters and Their Purpose

4.2 Challenges Encountered

Several challenges may occur during implementation. Water-quality datasets can contain **missing values, duplicate records, inconsistent formats, and different measurement scales**. These issues can affect the accuracy of the analysis. They can be addressed through data cleaning, appropriate handling of missing values, and standardization of numerical parameters.

Another challenge is combining water-quality and disease datasets because district names, time periods, and data formats may differ. Proper data matching and preprocessing are required

before performing correlation analysis. Selecting suitable parameters for PCA and interpreting the resulting components can also require careful analysis.

4.3 Possible Improvements

The current system can be improved by using larger and more recent datasets from multiple districts and time periods. Additional water-quality parameters and disease categories can also be included to improve the analysis.

The system can further be enhanced with **real-time water-quality data**, interactive maps, automated alerts, and improved visualization. More advanced statistical or predictive techniques can also be explored for identifying future risk patterns.

4.4 Recommendations

The proposed system can be recommended as a **supporting analytical tool** for water-quality monitoring and research. Regular collection of reliable water-quality and disease data can improve the quality of the results. Future work should include data from multiple years to identify seasonal and long-term patterns.

Further research can focus on validating the QRI against established water-quality standards and comparing results across different regions. The system should be used for **risk assessment and monitoring support**, while medical and public-health decisions should be made by qualified authorities.

Overall, the proposed system provides a useful foundation for **district-wise water-quality risk assessment, disease-pattern analysis, and data-driven decision support**.

CHAPTER 5

REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

5.1 Key Learning Outcomes

5.1.1 Academic Knowledge

The capstone project provided an opportunity to apply the concepts learned during the Information Technology course to a real-world environmental and public-health problem. The project improved the understanding of **Data Science, Machine Learning, Statistics, Principal Component Analysis (PCA), data preprocessing, correlation analysis, and data visualization.**

One of the major concepts learned was PCA, which was used to reduce multiple water-quality parameters into a smaller number of significant components. This helped in understanding how statistical techniques can be used to identify the most influential factors affecting drinking-water quality.

The project also strengthened the understanding of statistical correlation. Correlation analysis was used to study the relationship between the calculated water-quality risk index and reported disease cases. This helped in understanding how data-driven techniques can support decision-making in public-health and environmental applications.

5.1.2 Technical Skills

Several technical skills were developed during the project. These included:

- Python programming for data processing and analysis.
- Pandas and NumPy for handling and manipulating datasets.
- Scikit-learn for implementing PCA and data preprocessing.
- Statistical techniques for correlation analysis.
- Matplotlib and other visualization tools for presenting results.
- Data cleaning, normalization, and handling of missing values.
- Generating charts, risk classifications, and analytical reports.

5.1.3 Problem-Solving and Critical Thinking

The project involved several challenges related to data quality, missing values, different measurement scales, and interpretation of multiple water-quality parameters. These problems required careful analysis before applying PCA.

Standardization was necessary because the water-quality parameters had different units and ranges. Missing and inconsistent data also had to be identified and handled appropriately.

Another important learning was that statistical results should not be interpreted without considering their real-world context. For example, a correlation between water-quality risk and disease cases indicates an association but does not necessarily prove direct causation. This improved the ability to critically evaluate analytical results instead of relying only on numerical outputs.

5.2 Challenges Encountered and Overcome

5.2.1 Personal and Professional Growth

During the project, several difficulties were encountered while collecting, preprocessing, analyzing, and visualizing the data. Initially, understanding PCA and selecting suitable water-quality parameters was challenging.

There were also difficulties in handling missing or inconsistent values and ensuring that the dataset was suitable for statistical analysis. These challenges required repeated testing and modification of the data-processing steps.

At certain stages, errors in programming and unexpected analytical results caused frustration. However, debugging the code, studying the concepts again, and testing the solution step by step helped overcome these difficulties.

The experience improved patience, confidence, time management, and the ability to work independently. It also demonstrated that complex technical problems can be solved effectively through systematic analysis.

5.2.2 Collaboration and Communication

The project also provided experience in communicating technical ideas and discussing project decisions with teammates and supervisors. Different ideas were considered regarding

data selection, PCA implementation, risk classification, visualization, and presentation of results.

Coordination was sometimes challenging because team members had different approaches and schedules. These issues were addressed through regular discussions, task distribution, and sharing of progress.

The experience highlighted the importance of clear communication, responsibility, cooperation, and timely completion of assigned tasks. It also helped develop leadership and teamwork skills.

5.3 Application of Engineering Standards

Engineering principles and best practices were considered throughout the development of the project. Proper data preprocessing, systematic analysis, validation of results, and clear documentation were followed to improve the reliability of the system.

The project also emphasized:

- Structured software development.
- Proper data handling and validation.
- Reproducible analysis.
- Appropriate statistical methods.
- Clear visualization of analytical results.
- Responsible interpretation of health-related information.
- Protection of data integrity.
- Proper documentation of methodologies and results.

Following these practices helped produce a more organized, reliable, and understandable analytical system.

5.4 Insights into the Industry

The project provided a better understanding of how data-driven technologies are used to address real-world problems. In industries such as environmental monitoring, healthcare,

government services, and data analytics, large datasets are processed to support decision-making.

The project demonstrated that technical knowledge alone is not sufficient. Professionals must also understand data quality, domain requirements, communication, documentation, and ethical considerations.

The experience also increased interest in **Data Science, Machine Learning, and frontend/dashboard development**, as these areas provide opportunities to transform complex analytical results into useful and understandable information for users.

5.5 Conclusion of Personal Development

Overall, the capstone project contributed significantly to personal, academic, and professional development. It provided practical experience in applying theoretical concepts to a real-world problem involving drinking-water quality and disease-risk analysis.

The project improved programming, statistical analysis, problem-solving, teamwork, communication, and presentation skills. It also increased confidence in working with real datasets and analytical tools.

The experience has provided a stronger foundation for future studies and professional opportunities in **Information Technology, Data Science, Machine Learning, and Web-Based Data Visualization**.

CHAPTER 6

CONCLUSION

6.1 Conclusion

The **PCA-Based District-Wise Drinking Water Quality Risk Index and Disease Outbreak Correlation System for India** was developed to provide a systematic approach for analyzing drinking-water quality and identifying districts that may require greater attention.

The project addresses the problem of analyzing multiple water-quality parameters that differ in scale, importance, and characteristics. PCA was applied to reduce the dimensionality of the dataset and identify the major factors contributing to water-quality variation.

Based on the PCA results, a **Water Quality Risk Index** was generated for individual districts. The districts were then categorized into different risk levels, such as **Low, Moderate, High, and Very High Risk**. This provides a simpler way of understanding complex water-quality information.

The project further incorporated disease-outbreak information to examine the relationship between water-quality risk and reported water-related diseases. Correlation analysis provided an indication of whether districts with higher water-quality risk also showed higher levels of reported disease cases. The results are interpreted as associations rather than proof of direct causation.

The proposed system can support authorities, researchers, and public-health professionals by providing a data-driven overview of water-quality risk across districts. The visualization of risk levels and potential hotspots can assist in identifying areas that may require further investigation, monitoring, and preventive measures.

Overall, the project demonstrates how **PCA, statistical analysis, data processing, and visualization** can be combined to develop a useful decision-support system for environmental and public-health applications.

6.2 Project Significance

The major significance of the project is its ability to convert complex water-quality data into understandable district-level risk information. Instead of examining every water-quality parameter individually, the PCA-based approach identifies the most important patterns in the data.

The system can be further enhanced in the future by incorporating:

- Real-time water-quality data.
- More comprehensive disease datasets.
- GIS-based interactive district maps.
- Time-series analysis of water-quality changes.
- Machine-learning-based disease-risk prediction.
- Automated alerts for high-risk districts.
- Mobile-based access for field-level monitoring.

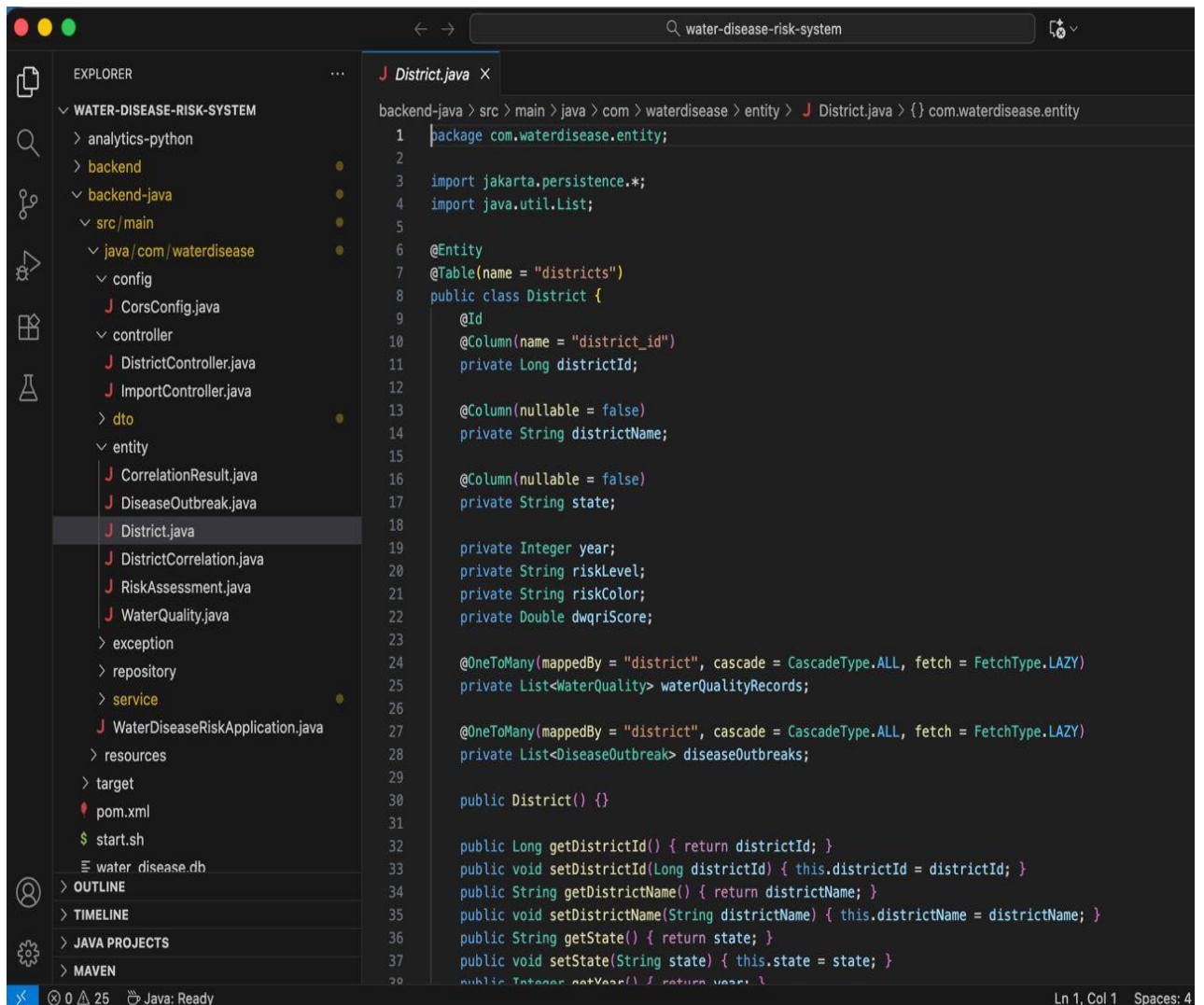
Therefore, the proposed system provides a foundation for developing a more advanced **data-driven water-quality and public-health monitoring platform for India**.

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APPENDICES

Code



The screenshot shows an IDE window with a project named "WATER-DISEASE-RISK-SYSTEM". The Explorer panel on the left shows the project structure, with the "entity" folder expanded, highlighting "District.java". The main editor displays the code for "District.java", which is a JPA entity class. The code includes imports for Jakarta Persistence and Java's List, and uses annotations like @Entity, @Table, @Id, @Column, @OneToOne, and @OneToMany. The class has private fields for districtId, districtName, state, year, riskLevel, riskColor, dwqriScore, waterQualityRecords, and diseaseOutbreaks, along with their corresponding getters and setters.

```
1 package com.waterdisease.entity;
2
3 import jakarta.persistence.*;
4 import java.util.List;
5
6 @Entity
7 @Table(name = "districts")
8 public class District {
9     @Id
10    @Column(name = "district_id")
11    private Long districtId;
12
13    @Column(nullable = false)
14    private String districtName;
15
16    @Column(nullable = false)
17    private String state;
18
19    private Integer year;
20    private String riskLevel;
21    private String riskColor;
22    private Double dwqriScore;
23
24    @OneToOne(mappedBy = "district", cascade = CascadeType.ALL, fetch = FetchType.LAZY)
25    private List<WaterQuality> waterQualityRecords;
26
27    @OneToOne(mappedBy = "district", cascade = CascadeType.ALL, fetch = FetchType.LAZY)
28    private List<DiseaseOutbreak> diseaseOutbreaks;
29
30    public District() {}
31
32    public Long getDistrictId() { return districtId; }
33    public void setDistrictId(Long districtId) { this.districtId = districtId; }
34    public String getDistrictName() { return districtName; }
35    public void setDistrictName(String districtName) { this.districtName = districtName; }
36    public String getState() { return state; }
37    public void setState(String state) { this.state = state; }
38    public Integer getYear() { return year; }
```

Output

